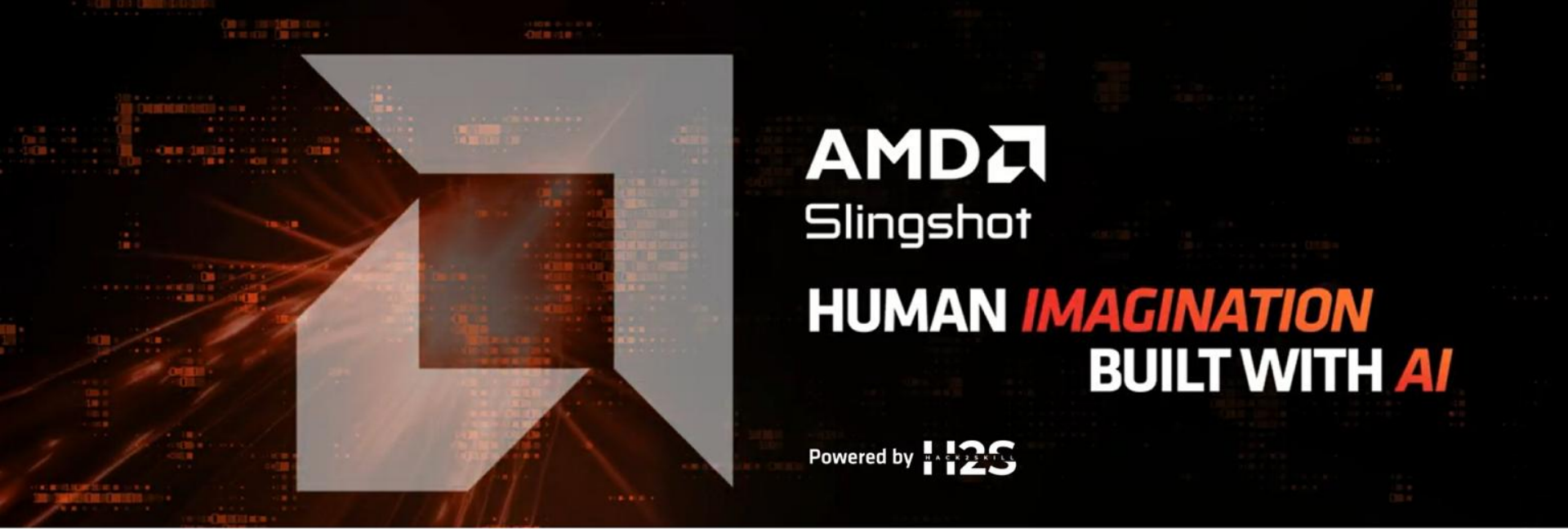




Team Details

- Team name – Tech Thrills
- Team leader name – Kabilan M
- Problem Statement - **AI in Education & Skilling** (Design and Development of Cognivara: An AI-Powered Educational Intelligence Platform for Deep Learning and Academic Integrity)

Brief about the idea

MAIN PROBLEM

Instead of learning, students are **receiving answers**. Online learning environments of today are designed for ease of use rather than understanding. Rather than teaching students how to think, AI tutors provide answers. Because AI **forgets** past conversations, learning becomes fragmented and each session begins anew. Pupils **who copy answers without understanding** pass tests without learning anything. Responses are generic and lack personalization. The worst part is that platforms are **unable** to determine whether a student has understood the AI or has simply repeated what it said. **The outcome?** Instead of producing critical thinkers, we are producing copy-pasters.

PROPOSED SOLUTION

Imagine it as your own AI tutor who really remembers you and cares about your learning. It asks **thought-provoking** questions instead of just giving answers. Instead of copying and pasting, students show their **understanding** of concepts by explaining them out loud. To help knowledge really stick, the system **plans** smart reviews, **recalls** your entire learning journey over several months, and explains complex topics with relatable examples in **multiple languages**.

INNOVATION & UNIQUENESS

- ✓ Our **persistent memory** remembers everything – your struggles, wins, and even your tone. Unlike ChatGPT that resets every chat, we're the tutor who actually knows you. No repeating yourself ever, like prompt again & again.
- ✓ There are many AI agents working together. There is an AI that teaches, an AI that checks understanding, and an AI that schedules. **It is better to have many agents working together** than to have a single bot trying to do everything.
- ✓ The student has to explain orally what he/she has learned. If you cannot explain it, then you have not learned it. It is as simple as that.
- ✓ The complex concepts are being translated into **Multiple languages**, along with examples that are culturally relevant. It is only when education is actually accessible that it is considered to be so.
- ✓ The **Oracle AI Database** are used for quick memory recall of thousands of students. It is a combination of enterprise-grade infrastructure that is tailored to education.

Opportunities

How different is it from any of the other existing ideas?

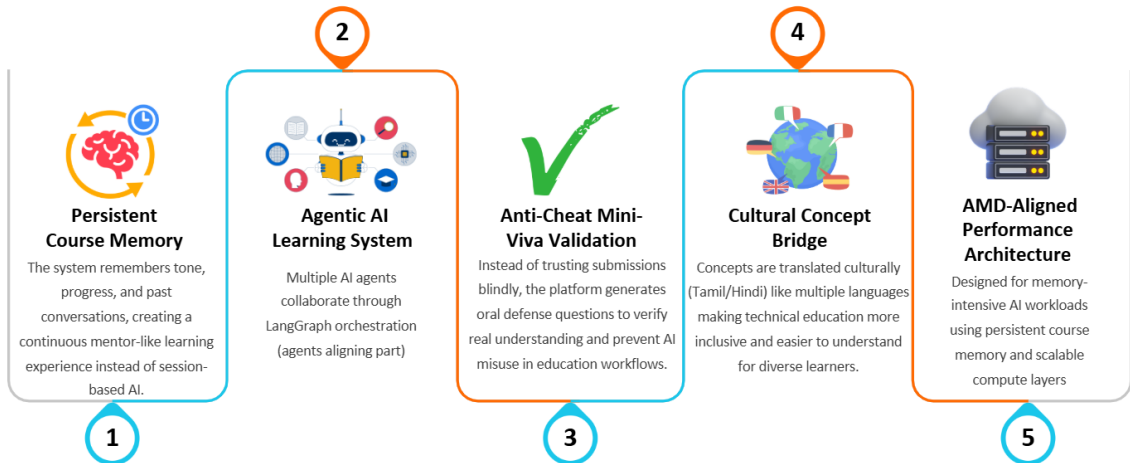
Existing Platforms Ask: "How can AI help students finish assignments faster?"

We Ask: "How can AI make students genuinely smarter?"

"What Others Do vs. What We Do"

Existing Product	Our Product
Build smarter answer machines	Build smarter students
Optimize for completion speed	Optimize for retention depth
Session-based interactions	Relationship-based learning
One AI does everything poorly	Specialized AI team engines
English-first, translate later	Culture-first, language follows

USP - User Selling Proposition



How will it be able to solve the problem?

Traditional Learning = Linear Growth

Study 10 hours | Retain 10 concepts.
Forget 7 | **Net gain: 3**

Our Learning = Exponential Retention

Study 10 hours | Retain 10 concepts.
Review 7 | Retain 9 | **Net gain: 9**

Answer Junkies

We make learning easier than copying.
Addiction gone in 14 days.

Zero Proof

Students defend their work verbally. Faking becomes impossible.

Memory Reset

AI tracks what you'll forget. Reviews lock it in permanently.

List of features offered by the solution

Socratic AI Mentor

Why?
What if?
How?

Stops handing you answers, starts making you think. Every question it asks builds a concept you'll never forget.

Oral Defense Validator

COMPREHENSION: 89%

Submitting work is easy. Explaining it? That's where real learning is proven. If you understood it, you can say it. If you can't, you didn't.

Persistent Memory

Remembers your entire journey. No 'start from scratch' every session.

Most AI forgets you the second you close the tab. Ours remembers every struggle, every breakthrough, every conversation.

Your entire learning journey - stored, tracked, and built upon every single session.

Smart Review Scheduler



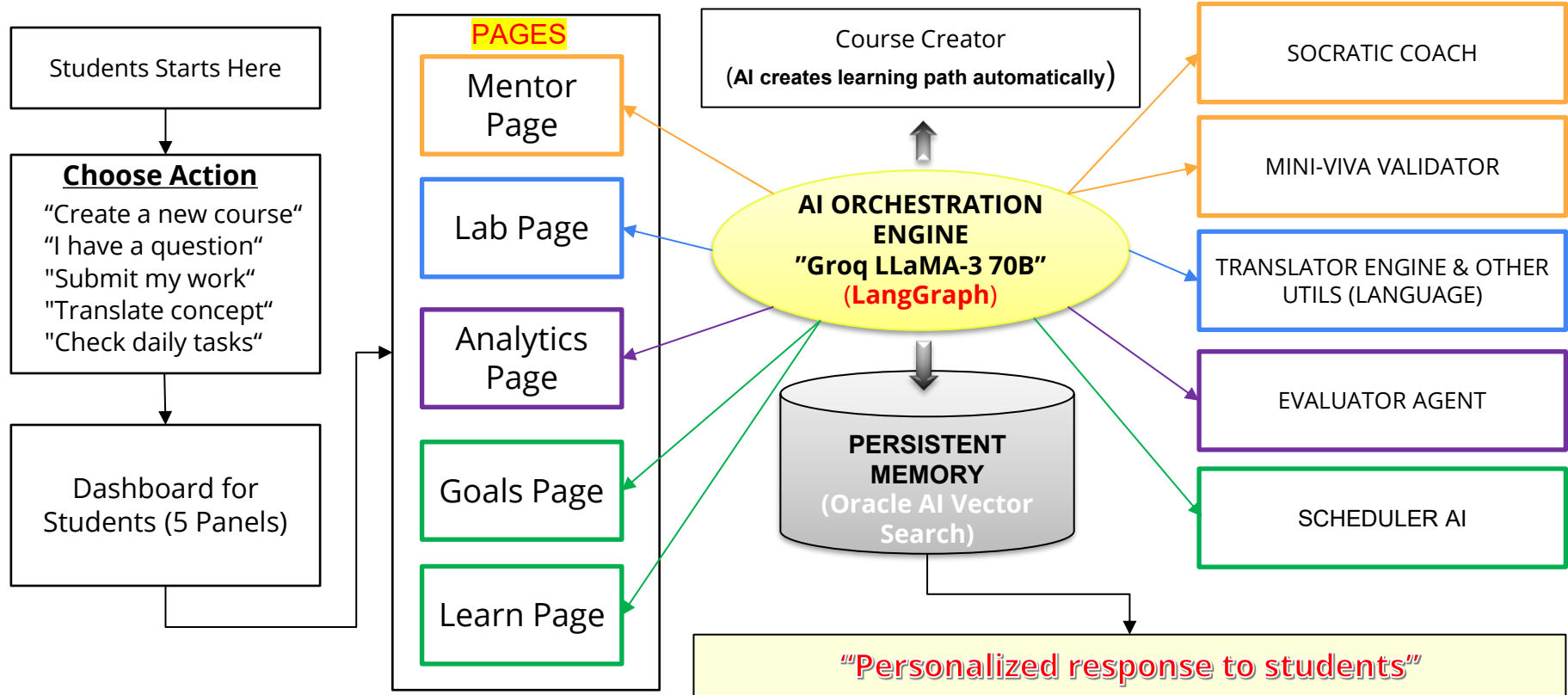
Forgetting is natural, we just won't let it happen. We review you exactly before you forget. Knowledge stays. Forever.

Cultural Concept Bridge

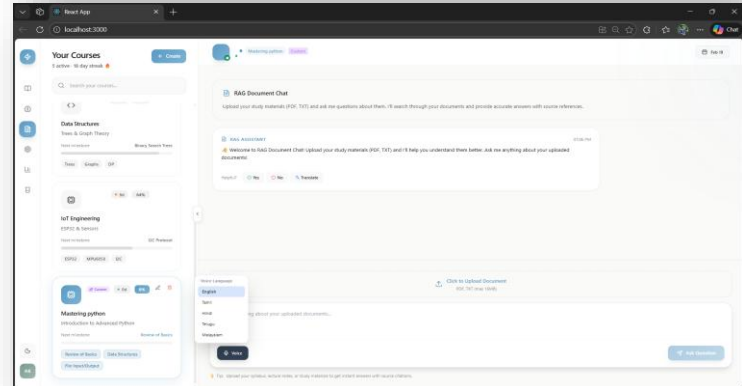
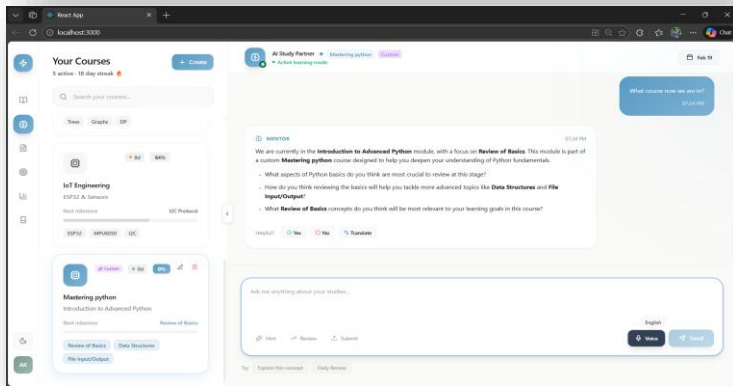
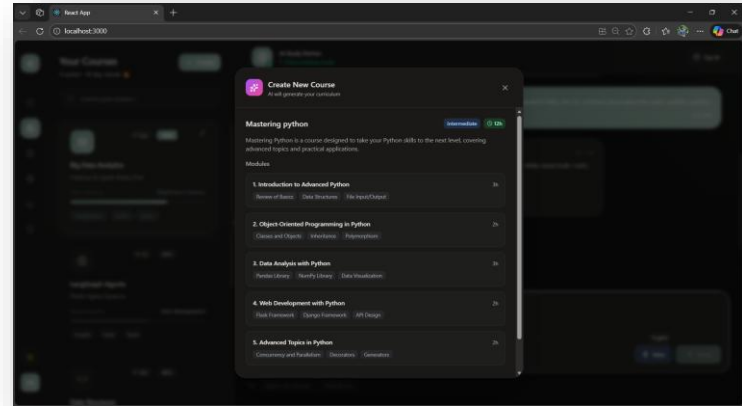
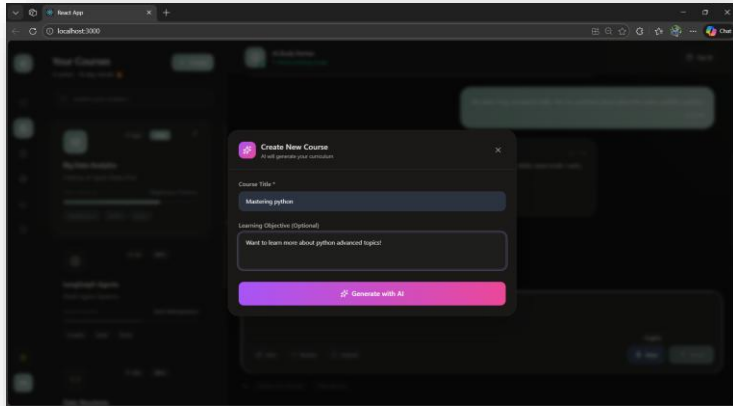
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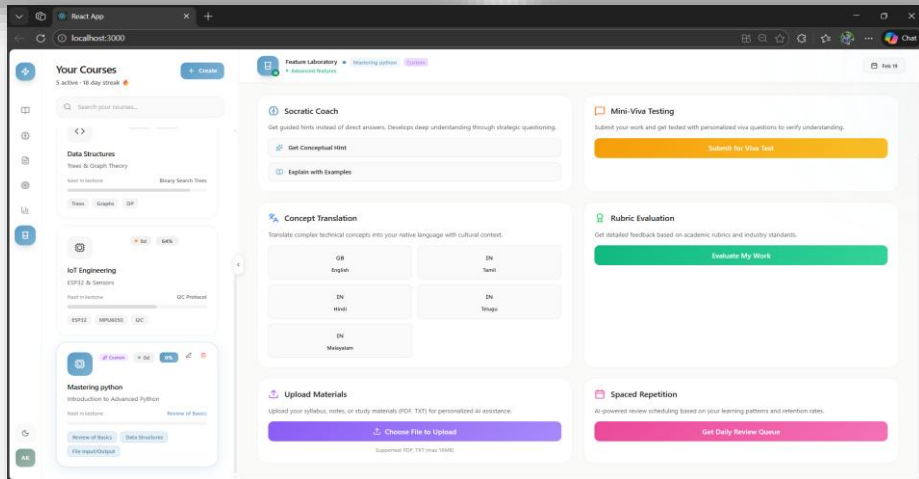
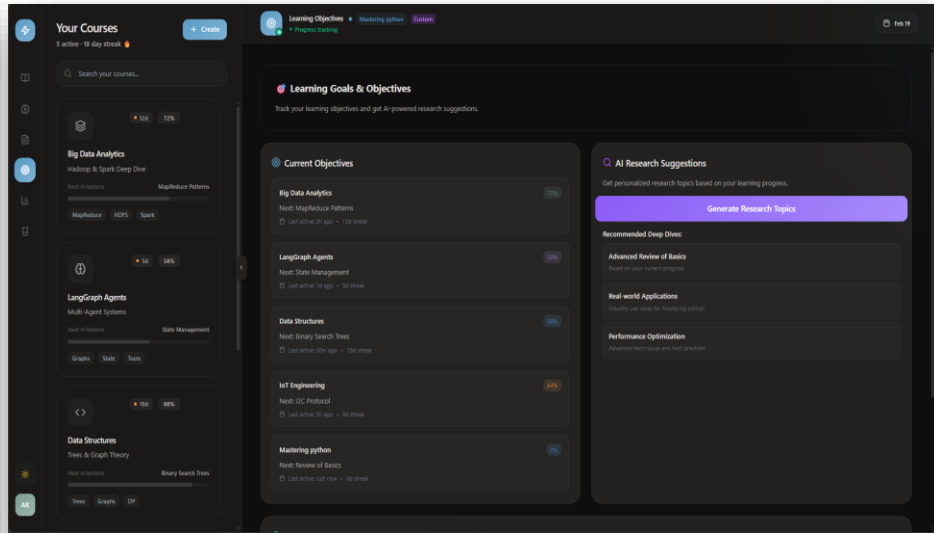
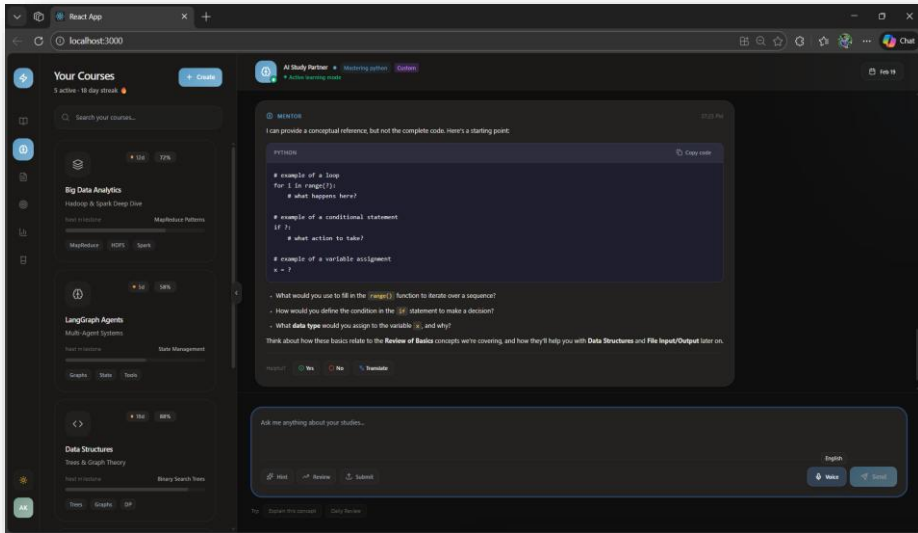
Tech makes more sense in your own language. We explain it the way your culture thinks, not just the way textbooks write.

Process flow diagram or Use-case diagram

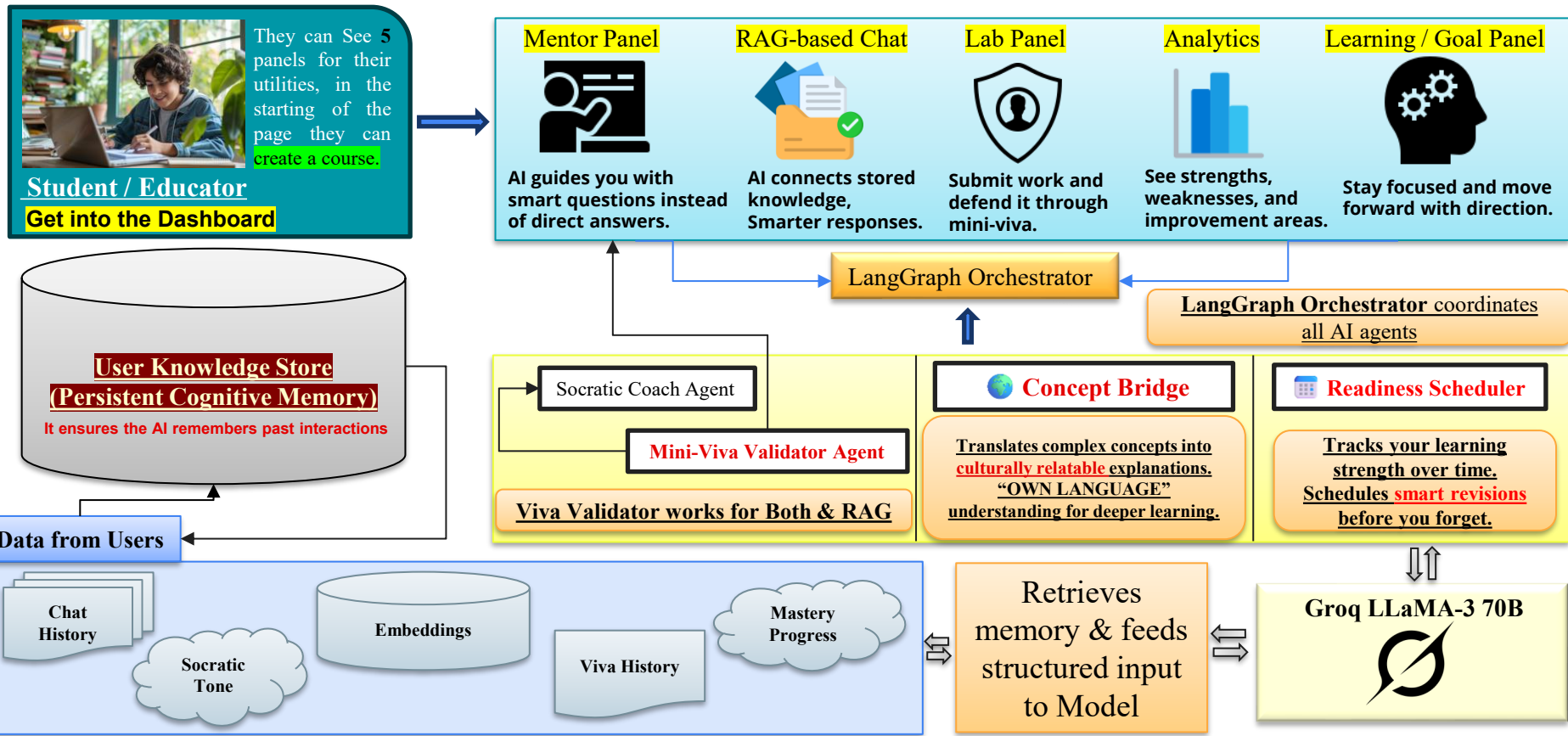


Wireframes/Mock diagrams of the proposed solution (optional)





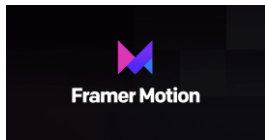
Architecture diagram of the proposed solution



Technologies to be used in the solution

Frontend Technologies

- ❖ **React 19.2.4** - Modern UI with component-based architecture
- ❖ **Framer Motion** - Smooth animations for educational interfaces
- ❖ **Lucide React** - Clean iconography for intuitive user experience
- ❖ **TailwindCSS** - Responsive design for multi-device learning



Backend & AI Infrastructure

- ❖ **Python Flask & FastAPI Server** - RESTful API orchestration
- ❖ **LangGraph Orchestrator** - Multi-agent AI workflow management
- ❖ **Groq LLaMA-3 70B** - High-performance inference engine
- ❖ **FAISS Vector Store** - Student-specific knowledge indexing



LangGraph



Usage of AMD Products/Solutions

AMD Instinct™ MI300X Accelerators

192GB HBM3 Memory - In-memory vector storage for instant similarity matching

Mixed-precision AI Training - Optimizing educational model fine-tuning

High-bandwidth Processing - Real-time inference for 1000+ concurrent students

Vector Embedding Acceleration - 1536-dimensional educational context vectors

AMD EPYC™ Processors

Database Optimization - Oracle 23ai query acceleration for student memory retrieval

Multi-agent Orchestration - Parallel processing of Socratic coaching workflows

Cache-efficient Architecture - Optimized similarity calculations for educational matching

Enterprise Security - Protecting student data and academic integrity

AMD ROCm Platform Integration

Open-source AI Stack - Seamless integration with educational AI frameworks

Custom Kernel Optimization - Educational-specific AI inference acceleration

Memory Management - Efficient handling of persistent student learning contexts

Multi-GPU Scaling - Supporting institution-wide deployment

Estimated implementation cost (optional)

1. Cloud Infrastructure To reduce capital expenditure, the MVP will use **cloud-based infrastructure** instead of purchasing physical GPUs or servers.

- ❖ Application Hosting (AWS): ₹5,000/month
- ❖ Managed Database (PostgreSQL , Firebase): ₹8,000/month
- ❖ Storage & Bandwidth: ₹5,000/month
- ❖ Monitoring & DevOps Tools: ₹3,000/month

Total Monthly Infra Cost: ₹20,000

6-Month Infra Cost: ₹1,20,000

2. LLM API & AI Services

Since the system uses API-based inference (**Groq/OpenAI**) and does not train foundation models, hardware GPUs are not required.

- ❖ LLM API Usage (6 months, estimated 500–1000 users): ₹1,00,000
- ❖ AI Embeddings & Processing Buffer: Included in above estimate

Total AI Cost (6 Months): ₹1,00,000

3. Development Team (Lean Structure)

For MVP, a compact and **efficient team** is sufficient:

- ❖ AI Engineer (6 months): ₹6–8 Lakhs
- ❖ Full-Stack Developer (6 months): ₹5–7 Lakhs
- ❖ UI/UX Support (Contract-based): ₹2 Lakhs

Estimated Development Cost: ₹15–18 Lakhs

4. Miscellaneous & Contingency

- ❖ Testing & Security Setup
- ❖ Domain & Deployment
- ❖ Legal / Documentation
- ❖ Buffer Reserve

Estimated Buffer: ₹2 Lakhs

Prototype Assets (Optional)

[GitHub Public Repository Link](#)

[kabilkforge06/Educational_Platform_AMD: AMD Hackathon 2026 | Next-Generation Adaptive Learning System with Anti-Cheat Intelligence](#) An innovative educational platform that revolutionizes online learning through AI-driven personalization, cultural awareness, and academic integrity verification. Built with React, Python, and advanced AI orchestration.

Add as per the requirements of the contest

Contest Alignment

- ✓ **Problem:** Online education lacks engagement and integrity verification
- ✓ **Solution:** AI-powered personalized learning with anti-cheat mechanisms
- ✓ **Innovation:** First multilingual Socratic coaching platform

AMD Integration

- ✓ **Hardware:** MI300X accelerators for real-time AI inference
- ✓ **Performance:** 192GB HBM3 memory for instant student context retrieval
- ✓ **Scalability:** EPYC processors for institution-wide deployment

Demo Status

- ✓ **Working Prototype:** Functional React + AI backend
- ✓ **Live Features:** Socratic coaching, oral defense validation
- ✓ **Performance:** Sub-200ms response times
- ✓ **Ready:** Containerized for immediate deployment

Key Differentiators

- ❖ **AMD-optimized:** Built specifically for Instinct™ hardware
- ❖ **Education-first:** Purpose-built for learning outcomes
- ❖ **Cultural intelligence:** Regional context adaptation
- ❖ **Integrity innovation:** Patent-pending anti-cheat system



AMD
Slingshot

HUMAN *IMAGINATION*
BUILT WITH *AI*

Powered by **I I2S**

Thank you!

